

# Altered exploratory behaviour and increased food intake in the spiny mouse before menstruation: a unique pre-clinical model for examining premenstrual syndrome

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Submitted on July 12, 2018; resubmitted on October 2, 2018; accepted on November 20, 2018

**STUDY QUESTION:** Does the newly discovered menstruating spiny mouse exhibit behavioural and metabolic changes in correlation with premenstrual phases of the menstrual cycle?

**SUMMARY ANSWER:** This is the first report of cycle variability in the exploratory and interactive behaviour, and food consumption in menstruating spiny mice, and demonstrates that physiological changes are also dependent on within-subject variation.

**WHAT IS KNOWN ALREADY:** Premenstrual syndrome (PMS) is a prominent cyclic disorder that affects millions of women worldwide. More than 70% of women endure symptoms of impending menstruation, such as bloating, abdominal cramping and nausea to some degree. Consequently, ~8% of women experience recurrent physical and emotional symptoms which are extreme enough to disrupt daily life and seek intervention. Due to a lack of an appropriate animal model, the mechanisms underlying PMS are poorly understood, and subsequently, effective treatments are limited.

**STUDY DESIGN, SIZE, DURATION:** This study analyses the changes in behavioural responses to the investigator during vaginal lavage ( $n = 14$ ), exploratory behaviour ( $n = 11$ ) and metabolism ( $n = 20$ ) across the menstrual cycle in the spiny mouse (*Acomys cahirinus*).

**PARTICIPANTS/MATERIALS, SETTING, METHODS:** We performed vaginal lavages on virgin spiny mice (6–8 months of age) and subjected each cohort of females to repeated measures for vaginal lavage, exploratory behaviour and metabolism. Stages of the menstrual cycle were designated as early follicular, late follicular, early luteal, late luteal, early menstrual and late menstrual, with the late luteal and early menstrual phases considered as premenstrual phases and analysed using generalized estimating equations. For vaginal lavage, the behavioural responses to researcher handling were scored on an increasing scale of severity during the lavage process (e.g. restraint, frequency of vocalizations, total handling time). For exploratory behaviour, exploration, memory and sociability were assessed through subjection to Open Field (OF), Novel Object Recognition (NORT), Social Novelty (SN) and Elevated Plus Maze (EPM) tests. For metabolism, physiological changes were measured over a 24-h period in metabolic cages. Results are mean  $\pm$  SD with statistical significance set to  $P < 0.05$ .

**MAIN RESULTS AND THE ROLE OF CHANCE:** Qualitative behavioural assessment showed that compared to early follicular controls, during premenstrual phases, cycling females had significantly increased probability of: manifesting difficulties during restraint (4X,  $P < 0.01$ ), vocalizing (8X,  $P < 0.01$ ) and exhibiting isolation in the cage (40X,  $P = 0.041$ ). We saw significant increases in handling time during the premenstrual phase in cycling females ( $76 \pm 16$  s) compared to controls ( $55 \pm 7$  s,  $P < 0.001$ ). For exploratory behaviour, cycling females in their early menstrual phase travelled significantly less distance in the outer zone of the OF arena ( $13.3 \pm 9.0$  m) than females in their early luteal phase ( $22.3 \pm 9.9$  m,  $P = 0.038$ ) and at significantly reduced velocities ( $40.2 \pm 10.5$  mm/s and  $78.8 \pm 31.0$  mm/s, respectively,  $P = 0.006$ ). These females also had fewer entries into the EPM open arms during the same phases ( $9.6 \pm 6.1$  and versus  $20.0 \pm 7.2$ , respectively,  $P =$

0.030) and travelled less distance ( $3.2 \pm 2.8$  m versus  $7.0 \pm 5.5$  m, respectively,  $P = 0.026$ ). No differences were observed in NORT or SN across the cycle. In the metabolism studies, spiny mice demonstrated a significant increase in food consumption (percentage of body weight) during the early follicular and late luteal phases ( $3.9 \pm 2.4\%$  and  $3.8 \pm 2.1\%$ , respectively) compared to the late follicular phase ( $2.3 \pm 2.6\%$ ,  $P = 0.015$ ).

**LARGE SCALE DATA:** N/A.

**LIMITATIONS, REASONS FOR CAUTION:** This is an observational study to determine fundamental changes in behaviour and metabolism in a novel species, and as such, lacks commercially available laboratory reagents and protocols specific to the spiny mouse.

**WIDER IMPLICATIONS OF THE FINDINGS:** The timing of these behavioural and physiological changes suggests that spiny mice exhibit symptoms analogous to PMS in higher order primates, thus providing a pre-clinical model for testing novel interventions to alleviate premenstrual symptoms and overcoming many limitations associated with this research area.

**STUDY FUNDING/COMPETING INTEREST(S):** N.B. is supported by a Research Training Program stipend through Monash University. J.E. is supported by a Fellowship awarded by the Peter Fielding Foundation. The Hudson Institute of Medical Research is supported by the Victorian Government Operational Research Infrastructure Support. The authors declare no conflicts of interest.

**Key words:** Menstruation / spiny mouse / premenstrual syndrome / behaviour / metabolism / rodent

## Introduction

The commencement of the menstrual cycle is identified by shedding of decidualised endometrium and accompanying overt vaginal bleeding in a non-conception cycle. Without the prolonged luteotrophic support provided by an embryo, a cascade of inflammatory events is mediated by the degeneration of the progesterone-secreting corpus luteum. Progesterone withdrawal positively regulates chemokine and proteolytic enzymes production, including interleukin-8 (Kelly *et al.*, 1994), matrix metalloproteases -3 and -9 (Salamonsen and Woolley, 1999; Salamonsen, 2003), and prostaglandins (Mannix, 2008; Martin *et al.*, 2014).

The intricacies of endometrial degradation and simultaneous repair are not entirely dissimilar to that of wound-healing (Salamonsen, 2003), involving complex interplay between ovarian and endometrial tissue, and the onset of menstruation is typically viewed as an inflammatory reaction. Menstrual onset prompts endometrial stromal oedema and upregulation of inflammatory mediators (Critchley *et al.*, 2001), with recent evidence even demonstrating cyclical changes in systemic inflammation, characterized by elevation in biomarker high sensitivity C-reactive protein during the early follicular phase as a consequence of menstruation (Blum *et al.*, 2005).

The range of predictable, cyclical symptoms during the late luteal and early menstrual phases, which subside with the onset of menstruation was first described as premenstrual tension in the early 1930s (Frank, 1931). This was later redefined as premenstrual syndrome (PMS) to encompass the diversity of both physical and behavioural symptoms, and the variation experienced by individuals (Greene and Dalton, 1953). Physical manifestations may be experienced in some degree by an estimated >70% of women worldwide, and include one, or combinations, of mastalgia, increased carbohydrate intake, hypertension, nausea, fatigue, sleep deprivation and abdominal pain (Woods *et al.*, 1982; Reid, 1991; Borenstein *et al.*, 2005; Mishell Jr, 2005; Gillings, 2014). One study also reports that symptoms are severe enough to disrupt functions of daily life in 8% of women (Yonkers *et al.*, 2008). Although research into the pathophysiology of PMS has recently progressed, formal criteria for specific diagnoses were only

proposed 30 years ago (Endicott, 2000). Reports of fluctuations in eating behaviour, sexual and social behaviour in wild yellow baboons (Hausfater and Skoblick, 1985) and captive vervet monkeys (Rapkin *et al.*, 1995) sparked newfound interest in the notion that there were potential biological and perhaps even evolutionary underlying foundations for the syndrome in higher order primates. Classical symptoms of PMS are also commonly linked with discernible behavioural changes; most notably, anxiety, anger/irritability, depression and social withdrawal, with anger often manifesting earlier than other emotional symptoms (Woods *et al.*, 1982; Ramcharan *et al.*, 1992; Meaden *et al.*, 2005; Yonkers *et al.*, 2008; Zaka and Mahmood, 2012).

The severity and subtypes of PMS are subject to individual variation, but previous work suggest that a significant contributor to functional impairment is mood (Bloch *et al.*, 1997). Dysphoric PMS, or premenstrual dysphoric disorder (PMDD) is considered a psychiatric subtype of PMS, demarcated by extreme disturbances in mood which averages levels reported in major depressive disorders and resulting in substantial social impairment (APA, 1997; Halbreich *et al.*, 2003; Gehlert *et al.*, 2009). PMDD has a current prevalence rate of 3–9%, though the true proportion of those afflicted is estimated to be higher at 13–18% (Halbreich *et al.*, 2003). In addition, both PMS and PMDD inflict a high economic burden, with an estimated 15% reduction of work productivity in the USA attributable to menstrual related problems, an indirect associated cost of up to \$4333 USD per patient (Borenstein *et al.*, 2005) and a calculated DALY (disability adjusted life years, equating to the number of healthy years lost to disability) of 4 years for PMDD patients (Halbreich *et al.*, 2003).

Despite many advances in identification and treatment of other menstrual conditions, the often debilitating effects of PMS and PMDD continue to be under-recognized. Although there are treatment options available for women, such as selective serotonin-reuptake inhibitors (SSRIs), GnRH agonists, and the oral contraceptive pill Yaz (drospirenone/ethinylloestradiol), to reduce the impact of physical symptoms, side effects (most commonly with SSRIs: insomnia, fatigue and gastrointestinal disruptions and Yaz: nausea, intermenstrual bleeding and headache) are frequently reported and may cause with women to withdraw from these treatments (Dimmock *et al.*, 2000; Pearlstein *et al.*, 2005; Yonkers *et al.*,

2005; Pearlstein and Steiner, 2012). In addition, the cost-efficiency and logistics of continuous versus intermittent treatment regimens of both anti-depressants and contraceptives may also impact patients.

The limitations imposed by the restricted number of menstruating species available for long-term research, having been confined to studies in humans and non-human primates, restricts rapid advancement in new potential therapies for premenstrual disorders. However, the discovery of a novel menstruating species belonging to the order Rodentia may play a key role in the future of menstrual disorder research. Our team identified a human-like menstrual cycle (lasting ~9 days) in the Egyptian spiny mouse (*Acomys cahirinus*), with similarities in pre-implantation decidualisation, progesterone profile and endometrial breakdown (Bellofiore et al., 2017).

Whilst we have shown the similarities in physiological aspects of menstruation between spiny mice, including early menstrual inflammation (Bellofiore et al., 2017, 2018b), behavioural changes corresponding with late luteal and early menstrual phases in this species have not yet been investigated. This is essential in determining the suitability of the spiny mouse as a complete and robust model of human-like menstrual biology. This study will examine whether behavioural and metabolic changes occur across the spiny mouse menstrual cycle, and propose the existence of anxiety-like behaviours during premenstrual phases (i.e. the late luteal and early menstrual phases).

## Materials and Methods

### Ethics

All research was approved in advance by the Monash University/Monash Medical Centre Animal Ethics Committee (#MMCB-2016/40), and all experimental procedures adhered to the Australian Code of Practice for the Care and Use of Animals for Scientific Purposes.

### Spiny mouse husbandry

The spiny mice used in this study were sourced from our in-house research colony (Monash University, Clayton, Australia). Virgin females aged 6–8 months and weighing 28–38 g were maintained as previously described (Bellofiore et al., 2017). Due to the fragility of spiny mouse skin, our animals are captured using a transparent, plastic container and are restrained with a small handtowel, as demonstrated in Fig. 1. Stages of the menstrual cycle were assigned as early follicular, late follicular, early luteal, late luteal, early menstrual and late menstrual, with the late luteal and early menstrual phases designated as being premenstrual. The average length of the menstrual phase is 3 days (Bellofiore et al., 2017), hence early menstrual was defined as within the first 24 h of blood in the vaginal cytology. Due to the susceptibility of spiny mice to demonstrate alterations in menstrual cycle length, and increased anxiety-like behaviour when introduced to foreign scents (i.e. unfamiliar people), a single investigator conducted all experimental procedures to reduce the risk of confounding the data. Prior to data collection, each cohort of females was acclimatized to the tests, which is specific to each and described in detail below. All parameters were assessed for the significance of repeated measures to account for individual subject variation, herein referred to interchangeably as 'subject matching' or 'subject pairing'.

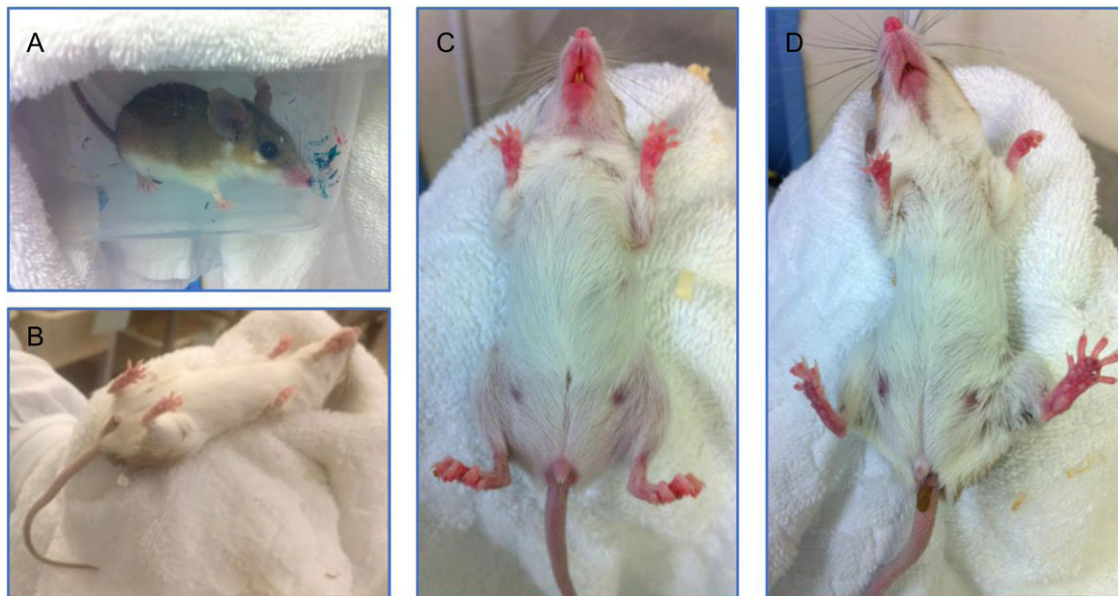
### Qualitative assessment of spiny mouse behaviour

To assess the changes in interaction between spiny mice and the researcher during routine vaginal lavage sampling,  $n = 14$  females were

scored on an increasing scale of severity of response (0 = None, 1 = Mild, 2 = Moderate, 3 = Severe) to 10 parameters: initial position in the cage prior to handling; weighing; towel; scruffing; restraint; lavage; abdominal palpation; vocalizations escapes; and return to cage (listed in detail in Table 1). Females were acclimated to the investigator, handling and behavioural suite for 4 days prior to collection of behavioural data. This is based on our own observations and previous determination of optimal acclimatization periods (unpublished observations). Spiny mice were subjected to daily vaginal lavage between 10:00 am and 12:00 pm and behavioural response scoring for 10 consecutive days (which we have previously proven to be well tolerated (Bellofiore et al., 2017)) to account for individual subject variation. During this assessment, the investigator was blind to menstrual cycle stage until completion of data collection for each female. At the end of every lavage process, a total stress score was calculated for each spiny mouse by combining the 10 scores for each above parameter, with the maximal total stress score being 30. At the end of 10 days, the scores were standardized for each individual by subtracting the lowest total stress score value from all other scores, e.g. if a female's lowest total stress score recorded was '7', and her highest total stress score was '23', her standardized total stress score would be '16'. In this way, changes in behaviour were relative to the individual female's baseline of normal behaviour. Other parameters measured included body weight (g), time taken to capture and overall procedure time (s). Data were analysed using repeated measures-ANOVA. Data were analysed using generalized estimating equations (GEE) to account for repeated measures in SPSS (IBM, Version 22, New York, NY, USA) by categorizing the spiny mice as having a 'Mild' (0 or 1) or 'Severe' (2 or 3) response. For GEE analysis, the early follicular phase served as the 'control' stage of the menstrual cycle, and data were deemed significant if changes  $P < 0.05$  observed versus this control phase.

### Quantitative assessment of spiny mouse behaviour

To quantify behavioural characteristics across the menstrual cycle,  $n = 11$  regularly cycling females were subjected to previously validated tests (Ratnayake et al., 2012). Supplementary Figure S1 demonstrates the behavioural arenas. All quantitative tests were conducted between 10:00 am and 4:00 pm. Spiny mice underwent blinded vaginal lavage immediately prior to testing. To ensure blind capture of each menstrual cycle stage, each female performed four tests within a 24 h period, with intermittent breaks across a 2-week period (Fig. 2A). In addition, the order of the tests was altered each time to minimize the risk of familiarization with the arenas. This protocol allows blinded capture of at least five of six menstrual cycle stages by a single investigator (necessary to minimize handling variation as outlined). After five sets of tests were completed, vaginal cytology was analysed to determine when a subject would enter the single remaining un-tested cycle phase to complete testing, and ensure cycle stages were not repeated and/or missed. As cycle synchronization upon co-housing does not occur in this species, the menstrual cycle stage of each female was variable. Hence, we consider the effect of order of tests taken/unblinding for only one stage to be negligible on data extraction. We trialled this sampling method using  $n = 5$  ovariectomised, age-matched control females and found no statistically significant effects of consecutive tests in any of the tests used herein (Supplementary Fig. S2). We therefore maintain the design of these behavioural tests are sufficient to remain blinded without evoking excess handling stress in the spiny mice and the results are unlikely to be confounded. Spiny mice were acclimated to the behavioural suite for a minimum of 30 min prior to testing. All behavioural tests were recorded using TopscanLite™ 2.0 (CleverSys, Reston, VA, USA), with GraphPad Prism 7.02 used to analysis statistically significant differences of  $P < 0.05$ . Data is presented as mean  $\pm$  SD.



**Figure 1** Methods for handling the spiny mouse (*Acomys cahirinus*). The capture (A) and handling (B) of spiny mice using a container and hand towel are shown. A placid female will lie flat in the supine position with limbs parallel to body (C); a more stressed female will try to twist head and body, limbs may be angled and spiny mouse may urinate or defecate (D).

#### Open Field (OF)

Females were placed in the centre of a  $50 \times 50 \text{ cm}^2$  square box with uniformly black walls 60 cm high under white light of 2.8 lux. Movement was recorded by a video camera for 5 min. Parameters assessed were time, distance and velocity in the inner and outer zones.

#### Novel Object Recognition Test (NORT)

This test was performed in two parts (Novel Object Recognition Test (NORT)1 and NORT2) with Open Field (OF) always preceding NORT1 and NORT2. In NORT1, two similar objects were placed in opposite corners of the OF arena (in this case, a 300 mL capacity square-based clear plastic bottles sealed with a metal sipper cap and 95 mm straight nozzle). The spiny mouse was then introduced and movement recorded for 10 min. The female was returned to her home cage for a minimum of 1 h before NORT2 commenced. One of the 'known' bottles was replaced by a slightly dissimilar 'unknown' object (a standard cylindrical glass beer bottle of 330 mL capacity, sealed with a metal bottle cap) before the spiny mouse was reintroduced into the arena for a further 10 min. The distance, velocity and periods of time spent exploring each of the objects was recorded. The discrimination index is presented as measure for object preference, on a scale from  $-1$  (known object 1) to  $1$  (known object 2 in NORT1 or unknown object in NORT2), with  $0$  being no preference. All bottles and the testing arena were cleaned thoroughly with ethanol and air dried between trials to remove potential olfactory cues. The bottles remained the same across each of the trials for the animals.

#### Social Novelty Test (SN)

The spiny mouse was placed into the central chamber of a plastic, open top rectangular box under standard room lighting (60 cm length  $\times$  30 cm width  $\times$  45 cm height) which was divided into three  $20 \text{ cm} \times 30 \text{ cm}$  chambers with spaces in the dividing walls to allow spiny mice to move between them. Females were allowed 5 min acclimatization to the box before she was removed. In each of the small holding boxes at opposite ends of the testing

arena, either a 'stranger' spiny mouse (of the same age and sex, but ovariectomized, see Supplementary Data) or a 'known' cage mate (part of experimental cohort in early follicular phase) was placed. The test subject was then reintroduced into the middle chamber of the box, and left to explore for 5 min, with the exploration of each of the three chambers recorded. The discrimination index is presented as measure for animal preference, on a scale from  $-1$  (known) to  $1$  (Stranger), with  $0$  being no preference. All holding boxes and the testing arena were cleaned thoroughly with ethanol and air dried between trials to remove potential olfactory cues. The known and stranger spiny mice were alternated across each of the trials.

#### Elevated Plus Maze Test (EPM)

The spiny mouse was placed at the intersection (10 cm  $\times$  10 cm platform 'central zone') of a wooden laminate elevated cross (50 cm above the ground) with four arms (40 cm length  $\times$  10 cm width); two arms were open and two enclosed by walls (20 cm height). The female's number of entries (defined as all four limbs within the specified region) in the open, closed or central zones, with distance and velocity travelled, were recorded for 5 min.

#### Metabolic cages

Females ( $n = 20$ ) were placed in glass cages for 24 h to measure food and water intake, and faeces and urine excretion (Supplementary Fig. S3). All spiny mice were trained and acclimatized to the cages for a period of 4 h the day before the experiment commenced. Semi-blind metabolic cage experiments were conducted in the time frame described in Fig. 2B. While similar to the behavioural testing timeline, metabolic cage isolation can increase the stress of the females if overly prolonged and subsequently alter females' menstrual cycles. Hence, this method of exposure to metabolic cage analysis has been optimized to account for adequate acclimatization to the cages and also provide sufficient rest periods without the need to re-train the females. The spiny mice were weighed and a vaginal lavage sample taken prior to isolation, and also weighed upon removal from the metabolic cage. All



**Table I** Semi-quantitative behavioural scoring of female spiny mice during vaginal lavage\*.

| Smearing                  | 0—Normal                                        | 1—Minor                                                                                   | 2—Moderate                                                                                                              | 3—High                                                                                                                         |
|---------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Initial position in cage  | Nesting with littermates With majority of group | Hiding in/under enrichment with 1–2 other littermates                                     | Hiding in/under enrichment alone OR Moderate Cage Climbing                                                              | Isolated, in cage corner farthest away from group OR Continual Cage Climbing                                                   |
| Response to weighing      | Still and placid                                | Initial movement (i.e. animal turns around 1–4 times) then placid                         | Animal constantly turning and/or attempts to escape                                                                     | Animal escapes one or more times                                                                                               |
| Response to towel         | Placid, no excessive movement                   | 1–2 repositions (i.e. animal will turn 180 degrees) Minor wriggling and/or foot scurrying | 3–4 repositions Repeated wriggling and/or foot scurrying Head/neck frequently exposed and tilted around container edges | Animal escapes from container one or more times Constant struggling/wriggling/scurrying Head/neck and torso frequently exposed |
| Ease of scuffing          | 1 attempt                                       | 2 attempts                                                                                | 3–4 attempts                                                                                                            | 5+ attempts                                                                                                                    |
| Attitude toward restraint | Placid, easily subdued, easily kept straight    | Exhibits slight apprehension to restraint, body may be at minor angle                     | Animal struggles, with claws stuck in towel fibres (hind or fore), minor kicking or head movement                       | Animal appears stressed, attempts to bite self or handler and repositioning does not alleviate                                 |
| Lavaging                  | Not bothered                                    | Some reaction, leg reflex at palpation                                                    | Intense hind body reaction, accompanying vocalization                                                                   | Total body reaction, all limbs flinch, torso movement, and/or animal attempts escape                                           |
| Abdomen palpation         | Not bothered                                    | Minor Flinch on one side (unilateral) LHS                                                 | Noticeable flinching, minor vocalization on one or both sides (uni or bilateral)                                        | Attempt to bite, noticeable vocalization                                                                                       |
| Vocalization              | None                                            | Rare 1–2                                                                                  | Frequent 3–5                                                                                                            | Constant 6+                                                                                                                    |
| Escaping                  | None                                            | 1 attempt (from container or cage) OR 1 escape from cage                                  | 2 or more attempts (from container and/or cage) OR 1 escape from container and one or more escapes from cage            | Multiple attempts from container and cage Escape upon returning to cage Escape on smearing                                     |
| Reaction to cage return   | Crawls out of restraint slowly, placid          | Small jump out of restraint, some initial hurried scurrying                               | Leap out of restraint, backflipping and cage climbing OR licking of lower abdomen                                       | Flip out of restraint, recapture necessary, constant backflipping and climbing on return +/- licking                           |

\*Females scoring a 0 or 1 were designated as having a ‘Mild’ response. Females scoring a 2 or 3 were designated as having a ‘Severe’ response

metabolic cage data are presented as change (+/–) in percentage (%) of initial body weight (bw) and data are presented as mean ± SD.

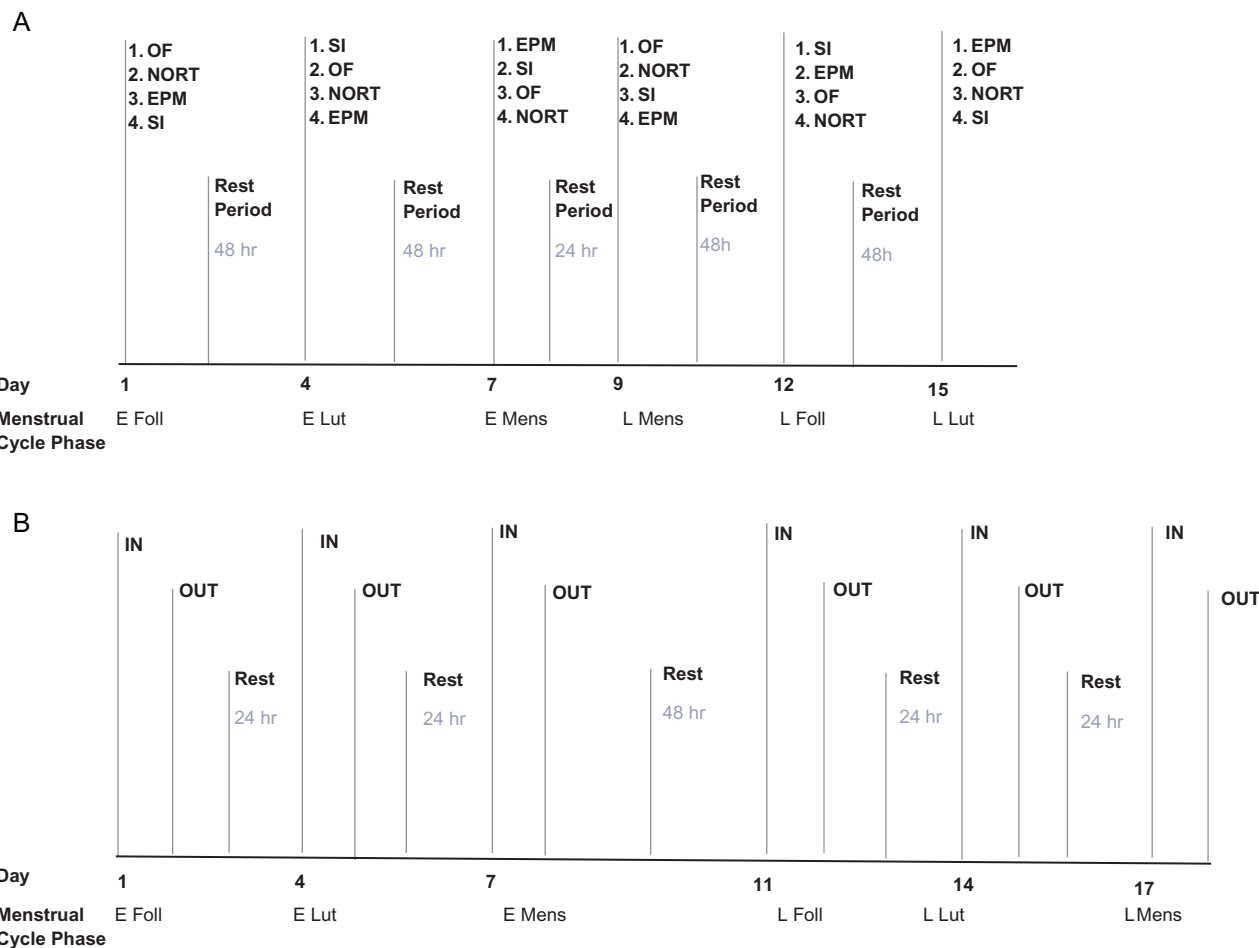
Results

Anxiety response in the spiny mouse across the menstrual cycle

Standardized behavioural scoring of female spiny mice demonstrated significant changes in response to multiple aspects of the vaginal lavage procedure during different phases of the menstrual cycle (Fig. 3 and Table II). We demonstrate that compared to the early follicular (EF) phase, early menstrual (EM) females were ~40 times more likely to be isolated ( $P = 0.041$ , Table II) and three times more likely to resist restraint ( $P = 0.012$ , Table II). Spiny mice in the late luteal (LL) phase were eight times more likely to frequently vocalize ( $P < 0.01$ ), and eight times more likely to respond severely to the handtowel holding during the LL phase ( $P < 0.01$ ), compared to the EF phase. The likelihood of females scoring severely upon return to the cage was increased 7-fold in early luteal (EL) females ( $P < 0.01$ ), and 2-fold in late menstrual (LM) females ( $P = 0.037$ ).

The overall stress score exhibited, established by combined responses to each parameter (initial position, weighing, towel, scruffing, restraint, lavage, abdominal palpation, vocalizations, escapes and return to cage) was increased in the LL phase ( $8.35 \pm 3.71$ ) compared to the EL phase ( $5.25 \pm 2.99$ ) (Fig. 3A). The total handling time of the spiny mouse significantly increased from  $62.5 \pm 9.9$  s in the EF phase to  $75.7 \pm 15.8$  s in the LL phase (Fig. 3B) reflecting the previously described stress/anxiety responses (Fig. 3A).

No difference was observed in time spent in either outer or central zones or the total distance travelled in the OF, although the effect of subject matching was significant in total distance travelled, outer zone distance and centre zone velocity (Fig. 4). Moreover, during the EM phase, females travelled significantly less distance in the OF outer zone ( $13.3 \pm 9.0$  m) than females in their EL phase ( $22.3 \pm 9.9$  m) and at significantly reduced velocities ( $40.2 \pm 10.5$  (EM) versus  $78.8 \pm 31.0$  mm/s (EL), respectively) (Fig. 4B and F). This behaviour was recapitulated in the Elevated Plus Maze Test (EPM) test (Fig. 5), whereby EM females travelled less distance in the open arms versus EL phase mice ( $3.2 \pm 2.8$  versus  $7.0 \pm 5.5$  m, respectively), with significantly reduced entries into the open arms ( $9.6 \pm 6.1$  (EM) versus  $20.0 \pm 7.2$  (EL)) and showed significantly reduced velocity through the central zone ( $42.5 \pm 22.6$  (EM)



**Figure 2** Timing of quantitative behavioural tests and metabolic cage experiments conducted on female spiny mice. (A) quantitative behavioural tests, (B) metabolic cage experiments. E = Early, L = Late, Foll = Follicular, Lut = Luteal phases of menstrual cycle. OF = Open Field, NORT = Novel object Recognition Test, EPM = Elevated Plus Maze, SN = Social Novelty behavioural tests. These protocols allow blind analysis if females exhibit a 9-day menstrual cycle exactly. However, due to natural variation in cycle length and phases, it is necessary to perform a lavage after (A) Day 12 and (B) Day 14 to estimate when the female will enter the final phase: (A) late luteal and (B) late menstrual. This is to ensure phases are not unnecessarily repeated or missed entirely. This may extend the time frame by up to 7 additional days depending on the length of the female's individual cycle.

versus  $84.9 \pm 50.0$  (EL) mm/s). The effect of subject matching was significant in open arm and central zone distance (Fig. 5B and C), open arm velocity (Fig. 5E) and number of closed arm entries (Fig. 5G).

No preference for the known or unknown object was observed in NORT2 between stages of the menstrual cycle (Fig. 6). However, in both NORT 1 and 2, a significant effect of matching was observed in both distance and velocity (Fig. 6B, C, E and F). No differences in social novelty were apparent across the menstrual cycle (Fig. 7A), although spiny mice showed preference to explore either of the other animals over the neutral zone in all phases (Fig. 7B).

## Metabolic changes in the spiny mouse across the menstrual cycle

Spiny mice demonstrated cyclical changes in food consumption (Fig. 8A), with a bimodal pattern exhibited. The period of greatest food intake occurs during the EF phase at  $3.9 \pm 2.5\%$  bw; prior to ovulation. The secondary peak occurs during the LL phase at  $3.8 \pm 2.1\%$

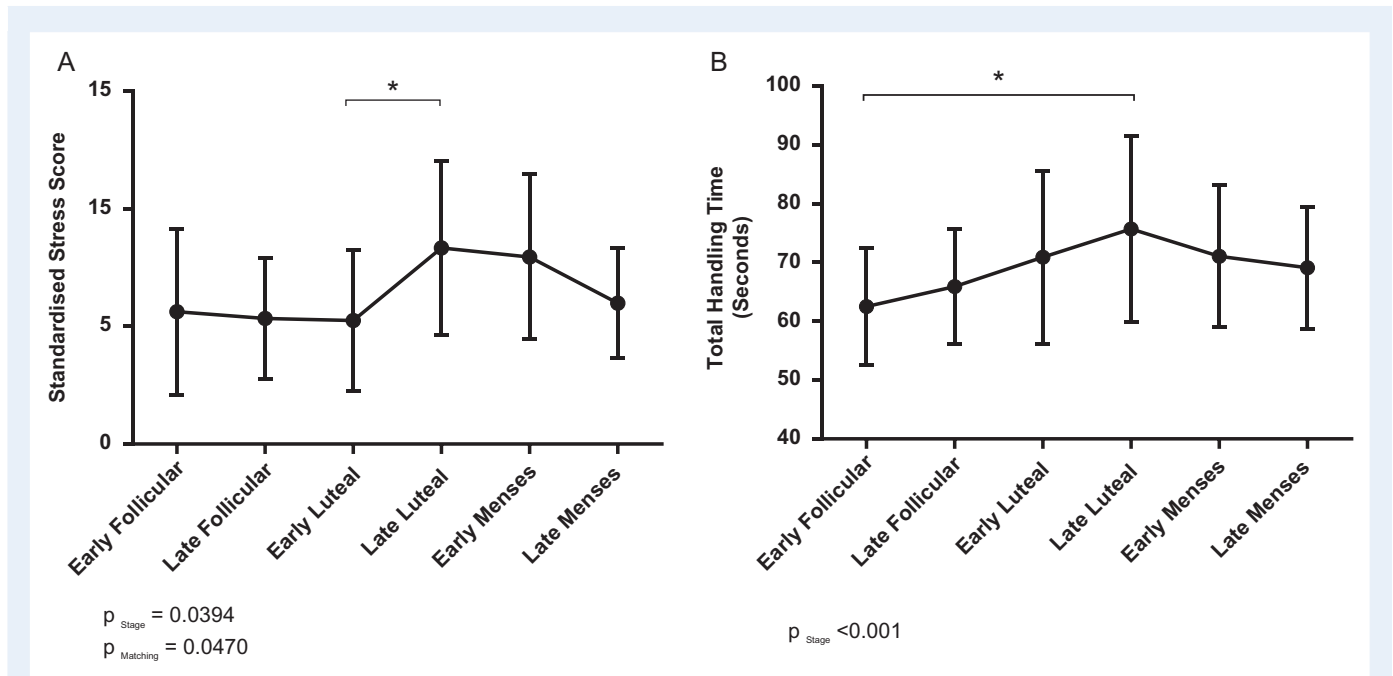
bw, preceding the onset of menstruation. No changes were observed between phases of the menstrual cycle for water consumption and urine excretion (Fig. 8B and C, respectively), although within-subject variation is a significant contributor to both results obtained.

Despite substantial increases in food intake, no significant changes in percentage of body weight upon removal from the metabolic cage were recorded (Fig. 8D). This is contrary to significant cyclical fluctuations in bw of spiny mice maintained in conventional cages (Fig. 8E), peaking during the LF phase ( $1.4 \pm 1.4\%$ ) and troughing during the EL phase ( $-1.5 \pm 2.0\%$ ).

The menstrual cycle with specific reference to the timing of phases comparing the spiny mouse and human and the onset of premenstrual symptoms is summarized in Fig. 9.

## Discussion

Our study is the first to report both behavioural and metabolic alterations across the menstrual cycle of a rodent species, which reflect



**Figure 3** Qualitative assessment of stress and handling in spiny mice. (A) Relative overall stress score and (B) overall handling time during vaginal lavage procedure of females ( $n = 14$ ) across the menstrual cycle. Significant results shown  $P < 0.05$  (\*), mean  $\pm$  SD, repeated measures ANOVA.

those observed in human females. This further reinforces the similarity of spiny mouse reproductive physiology and associated behaviours to that of menstruating humans and non-human primates. We also advance our understanding of the biology of this captive species as an emerging research model.

Of note, the menstrual cycles of females do not develop spontaneous pseudopregnancy when spiny mice are housed with several cage mates, and do not synchronize based on cage mates or the influence of an alpha-female as demonstrated by other rodents and non-human primates, respectively (Van der Lee and Boot, 1955; Parkes and Bruce, 1961; Rapkin Pollack et al., 1995). Therefore, each animal used in the current study was carefully staged, with the majority of females typically not in the same menstrual cycle stages as their cage mates. This staging is critical for interpretation of study data and is an essential finding for use of this unique model in future studies of menstrual physiology.

Behavioural scoring of individual females' interactions with the investigator demonstrated a reduced tolerance to the lavage process during premenstrual and early menstrual phases, as evidenced by a striking preference for isolation, increased frequency of vocalizations and resistance to the handtowel restraint, as well as an increase in overall handling time and in standardized stress scores. By accounting for the variation in individual subject behaviour, we are able to demonstrate relative changes not only in overall phases of menstrual cycle, but within the subject herself compared to her own baseline of 'normal'. Again, the careful mapping of each female is critical for understanding overall behavioural changes. As with humans, the response of each spiny mouse is unique and great care must be taken not to ascribe generalized changes to a study cohort. Primate studies demonstrate a preference for perimenstrual female baboons to eat in treetops and

avoid interaction with others (Hausfater and Skoblick, 1985) with initiation of avoidance and submission in vervet monkeys suggestive of behavioural characteristics analogous to PMS in humans (Rapkin et al., 1995). Despite spiny mice living as social animals demonstrating a tendency to huddle together, our observation of high preference for isolation during premenstrual phases suggests a similar preference for social withdrawal, further highlighting parallels with other menstruating species. Furthermore, this preference for isolation during the EM phase corresponds with previously recorded progesterone withdrawal in the spiny mouse (Bellofiore et al., 2017), providing support of a temporal correlation of progesterone withdrawal and the onset of mood-related premenstrual symptoms in this species (Reynolds et al., 2018), as demonstrated in rat models using exogenous hormonal manipulations (Li et al., 2012).

The initial semi-quantitative cyclical changes in behaviour were validated by OF and EPM tests; both designed to test anxiety and exploratory behaviour. During the OF, the time spent in the outer and centre zones did not differ across the cycle. However, a significant decline in the distance travelled in the outer zone in conjunction with significantly reduced velocity emphasizes the presence of altered locomotor function, which may indicate an unwillingness for extensive movement due to pain. However, coupled with no difference observed in number of closed arm entries, plus reduced distance and number of entries in the EPM open arms, this suggests that avoidance and anxiety are likely involved. In rodents, spontaneous (non-stimulus evoked) pain is a challenging parameter to measure and often subjective, but further investigation could clarify whether pain is an underlying factor in these observations. Grimace scales (facial expressions) and burrowing assays (Deuis et al., 2017) could potentially be assessed if validated in the spiny mouse.

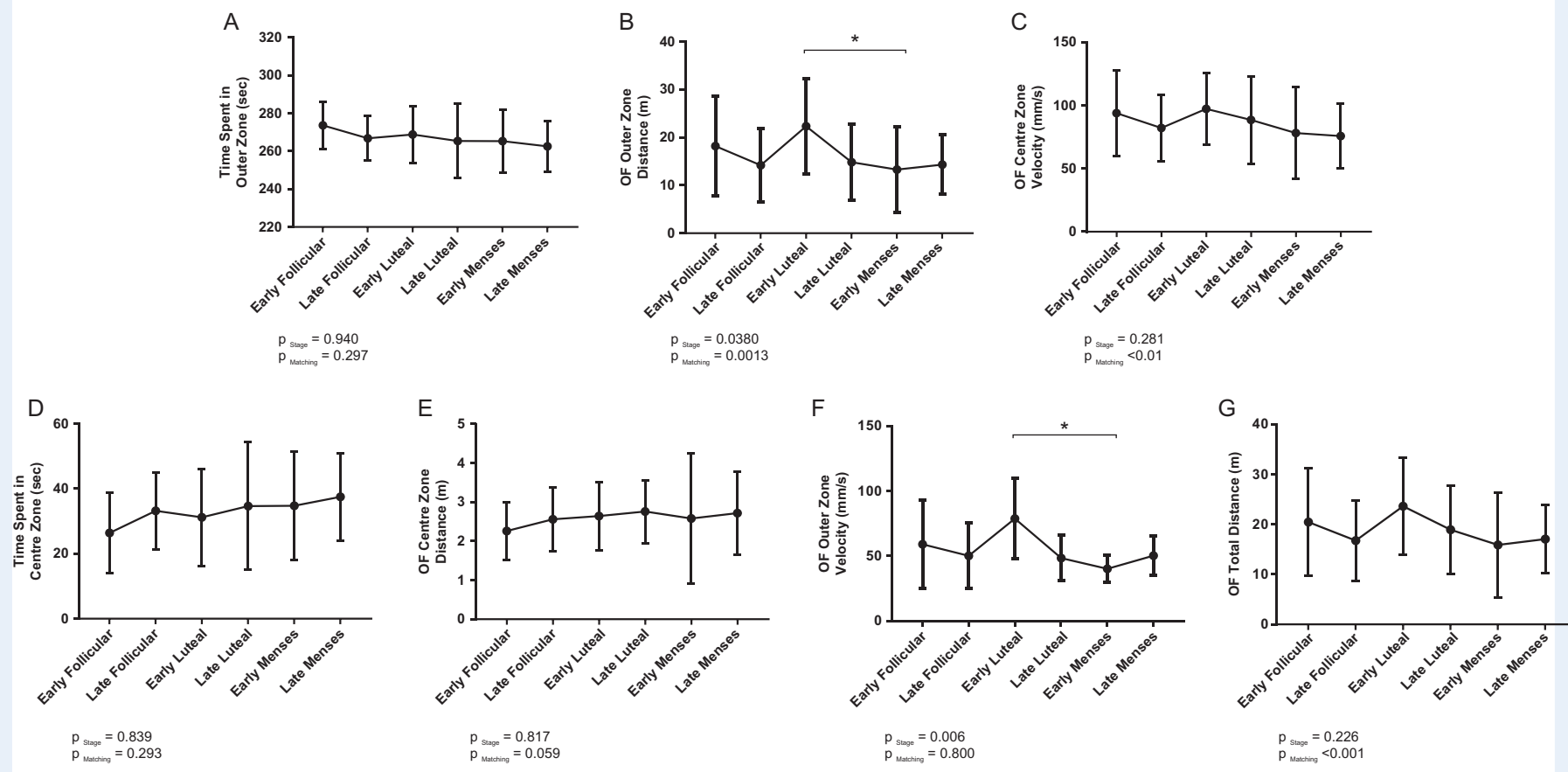
**Table II** Semi-quantitative behavioural scoring analysis of female spiny mice.

| Parameter           | Cycle stage                | Odds ratio (95% CI) | P value (* = significant) | Parameter     | Cycle stage                | Odds ratio (95% CI) | P value (* = significant) |
|---------------------|----------------------------|---------------------|---------------------------|---------------|----------------------------|---------------------|---------------------------|
| Initial position    | Early follicular (control) | 1.0                 | —                         | Weighing      | Early follicular (control) | 1.0                 | —                         |
|                     | Late follicular            | 7.3 (1.0–54.2)      | 0.932                     |               | Late follicular            | 0.6 (0.1–3.2)       | 0.587                     |
|                     | Early luteal               | 19.1 (5.0–72.8)     | 0.200                     |               | Early luteal               | 0.3 (0.0–1.7)       | 0.165                     |
|                     | Late luteal                | 9.4 (3.7–55.8)      | 0.388                     |               | Late luteal                | 0.9 (0.2–4.2)       | 0.883                     |
|                     | Early menstrual            | 40.7 (8.5–195.0)    | 0.041*                    |               | Early menstrual            | 0.4 (0.1–2.4)       | 0.338                     |
|                     | Late menstrual             | 10.5 (2.1–54.7)     | 0.736                     |               | Late menstrual             | 1.2 (0.3–5.6)       | 0.816                     |
| Towel               | Early follicular (control) | 1.0                 | —                         | Scruffing     | Early follicular (control) | 1.0                 | —                         |
|                     | Late follicular            | 4.4 (1.5–12.6)      | 0.03*                     |               | Late follicular            | 0.82 (0.1–7.3)      | 0.954                     |
|                     | Early luteal               | 1.3 (1.3–1.4)       | 0.02*                     |               | Early luteal               | 0.81 (0.3–2.4)      | 0.908                     |
|                     | Late luteal                | 8.4 (4.7–15.0)      | <0.01*                    |               | Late luteal                | 0.25 (0.03–1.9)     | 0.224                     |
|                     | Early menstrual            | 4.4 (4.1–4.7)       | <0.01*                    |               | Early menstrual            | 0.53 (0.3–0.9)      | 0.063                     |
|                     | Late menstrual             | 1.3 (0.1–25.4)      | 0.968                     |               | Late menstrual             | 0.53 (0.1–4.3)      | 0.623                     |
| Restraint           | Early follicular (control) | 1.0                 | —                         | Lavage        | Early follicular (control) | 1.0                 | —                         |
|                     | Late follicular            | 0.9 (0.2–4.0)       | 0.931                     |               | Late follicular            | 0.0 (0.0)           | 0.999                     |
|                     | Early luteal               | 1.1 (0.5–2.9)       | 0.637                     |               | Early luteal               | 0.54 (0.0–7.0)      | 0.647                     |
|                     | Late luteal                | 3.6 (1.2–10.4)      | <0.01*                    |               | Late luteal                | 0.43 (0.0–5.3)      | 0.508                     |
|                     | Early menstrual            | 3.6 (0.9–15.1)      | 0.063                     |               | Early menstrual            | 0.54 (0.0–7.0)      | 0.647                     |
|                     | Late menstrual             | 4.4 (1.1–20.4)      | 0.043*                    |               | Late menstrual             | 0.46 (0.0–5.8)      | 0.549                     |
| Abdominal palpation | Early follicular (control) | 1.0                 | —                         | Vocalizations | Early follicular (control) | 1.0                 | —                         |
|                     | Late follicular            | 0.5 (0.2–2.3)       | 0.395                     |               | Late follicular            | 4.4 (1.5–2.6)       | 0.03*                     |
|                     | Early luteal               | 0.5 (0.2–2.3)       | 0.395                     |               | Early luteal               | 1.3 (1.3–1.4)       | 0.02*                     |
|                     | Late luteal                | 0.5 (0.2–2.3)       | 0.395                     |               | Late luteal                | 8.4 (4.7–15.0)      | <0.01*                    |
|                     | Early menstrual            | 0.5 (0.2–2.3)       | 0.395                     |               | Early menstrual            | 4.4 (4.1–4.7)       | <0.01*                    |
|                     | Late menstrual             | 1.8 (0.5–6.9)       | 0.390                     |               | Late menstrual             | 1.3 (0.1–25.4)      | 0.968                     |
| Escapes             | Early follicular (control) | 1.0                 | —                         | Return        | Early follicular (control) | 1.0                 | —                         |
|                     | Late follicular            | 0.3 (0.0–1.7)       | 0.165                     |               | Late follicular            | 1.2 (0.3–4.0)       | 0.861                     |
|                     | Early luteal               | 0.9 (0.2–4.2)       | 0.883                     |               | Early luteal               | 7.0 (1.7–29.5)      | <0.01*                    |
|                     | Late luteal                | 1.2 (0.3–5.5)       | 0.816                     |               | Late luteal                | 1.2 (0.4–3.0)       | 0.824                     |
|                     | Early menstrual            | 0.3 (0.0–1.7)       | 0.165                     |               | Early menstrual            | 1.7 (0.4–6.5)       | 0.634                     |
|                     | Late menstrual             | 0.3 (0.0–1.7)       | 0.165                     |               | Late menstrual             | 1.9 (1.4–17.1)      | 0.037*                    |

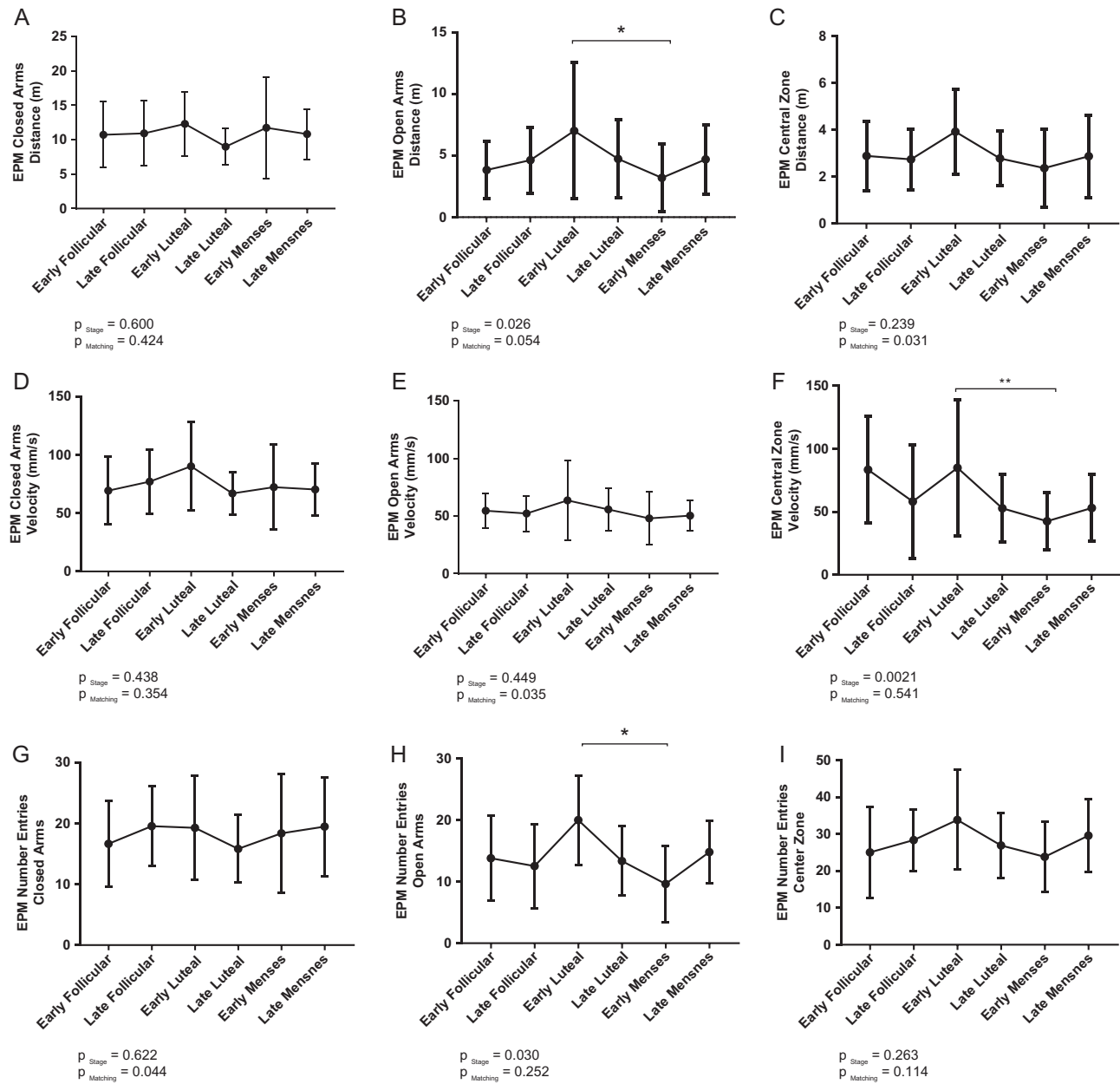
The spiny mice showed peak exploration, as depicted by distance travelled and velocity, during the early luteal phase, and a nadir in exploration during the EM phase correlating with the time of social isolation described above. In rats, females are typically most active during oestrous (Anderson, 1940; Birke and Archer, 1975), when oestrogen levels are highest and ovulation occurs. Our expectation was that peak exploration would occur during the late follicular phase, when vaginal cytology shows prominently cornified epithelial cells. However, the very final stages of oestrous may overlap with infiltration of leukocytes and there is a small possibility of cytology being deemed metestrous/early luteal phase thus there is a possibility in the current study that late follicular samples may have been misclassified as early luteal. Nonetheless, reduced physical activity in the spiny mouse may reflect

the premenstrual symptoms seen in women of fatigue, abdominal pain (possibly caused by bloating), anxiety associated with progesterone withdrawal (Sanders *et al.*, 1983; Gonda *et al.*, 2008; Reynolds *et al.*, 2018), or a combination of these. It is thought that the endocrinology of the spiny mouse likely mimics menstruating species more closely than oestrous species, with oestrogen levels rising in unison with progesterone during the luteal phase to support decidual cell proliferation. Hormone assays within the spiny mouse in conjunction with cytology would enable more accurate cycle staging. However, the precise levels of oestrogen and fluctuations throughout the menstrual cycle have not yet been determined in spiny mice due to the immense difficulty in validating methods of measurement, the high likelihood of cross reactivity when using commercially derived antibodies or ELISA kits, and the fact





**Figure 4** OF analysis across the menstrual cycle in female spiny mice. Outer zone time (A), distance (B) and velocity (C). Centre zone time, distance and velocity (D–F, respectively). Total distance travelled (G). Significant decrease in OF outer zone distance (B) and velocity (F) during early menses compared to the early luteal phase (\* $P < 0.05$ ). Significant effect of subject matching in B, C, E and G ( $n = 11$ ), mean  $\pm$  SD, Repeated measures-ANOVA.

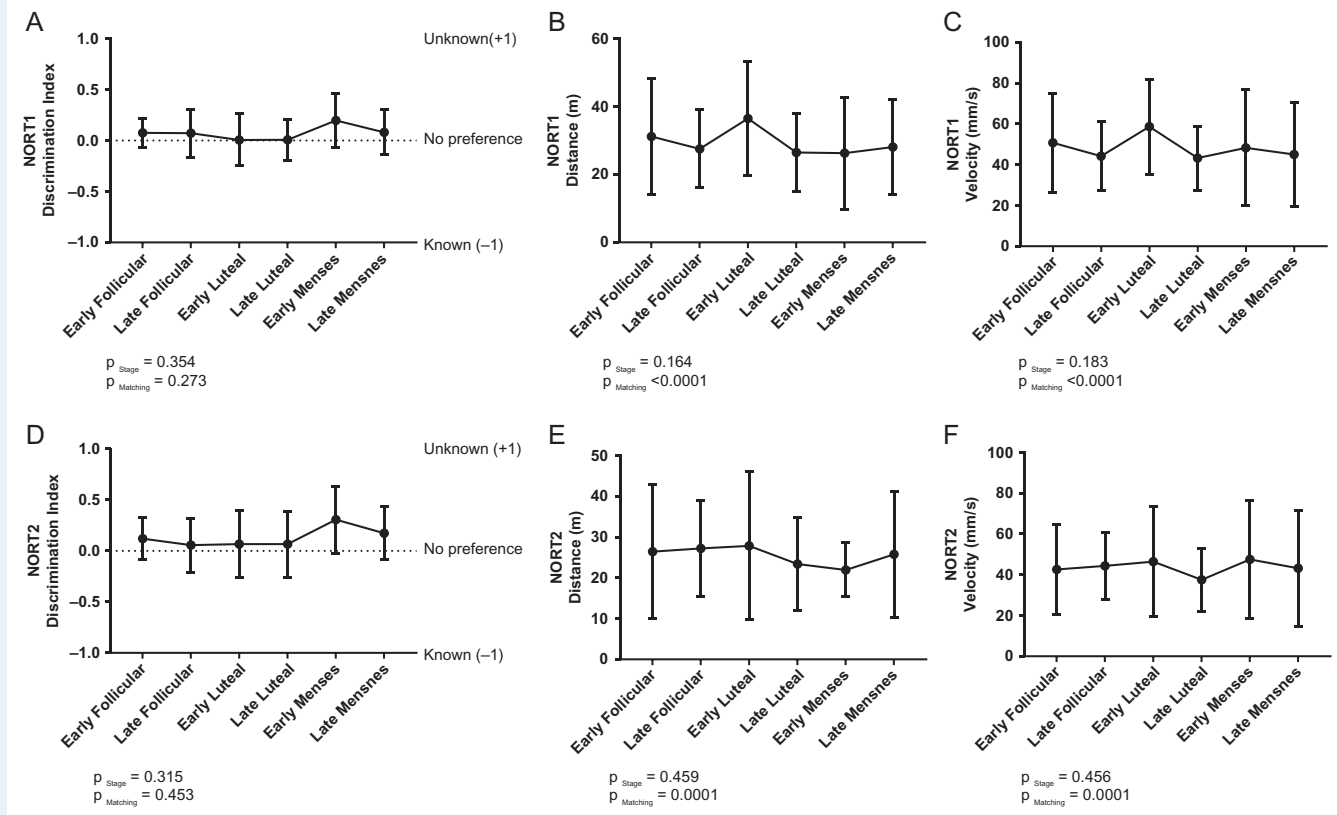


**Figure 5** EPM analysis across the menstrual cycle in female spiny mice. Distance travelled in EPM closed arms (A), open arms (B), and central zone (C), velocity (D–F) and number of entries in respective zones (G–I). Distance travelled in open arm (B), velocity in central zone (F) and number of entries in open arms show significant results (\* $P < 0.05$  and \*\* $P < 0.001$ ), ( $n = 11$ ), mean  $\pm$  SD, Repeated measures-ANOVA. Significant effect of subject matching present B, C, E and G.

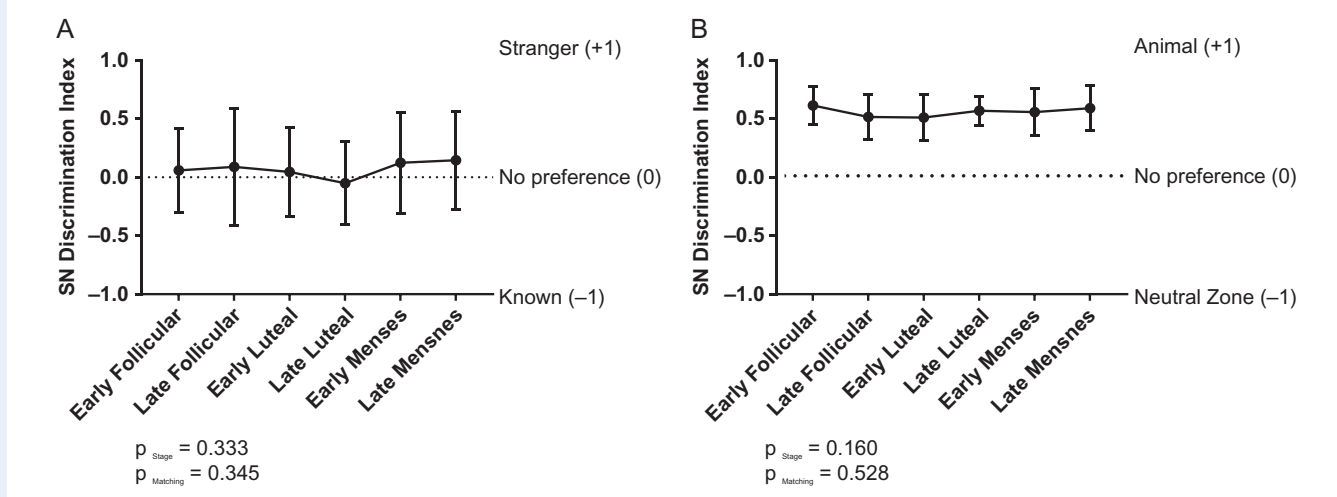
the spiny mice often behave as both human and rodent when seeking to determine gene or protein expression (Bellofiore *et al.*, 2018a,b). These obstacles are currently being investigated (Bellofiore *et al.*, 2017, 2018a, b).

Interestingly, despite the behavioural changes described above, no differences were observed when testing for memory or social novelty between phases of the menstrual cycle. Our social novelty test is a modified version of the previously used social interaction test in the spiny mouse (Ratnayake *et al.* 2012), where interaction with a known and stranger mouse was simultaneous to identify whether

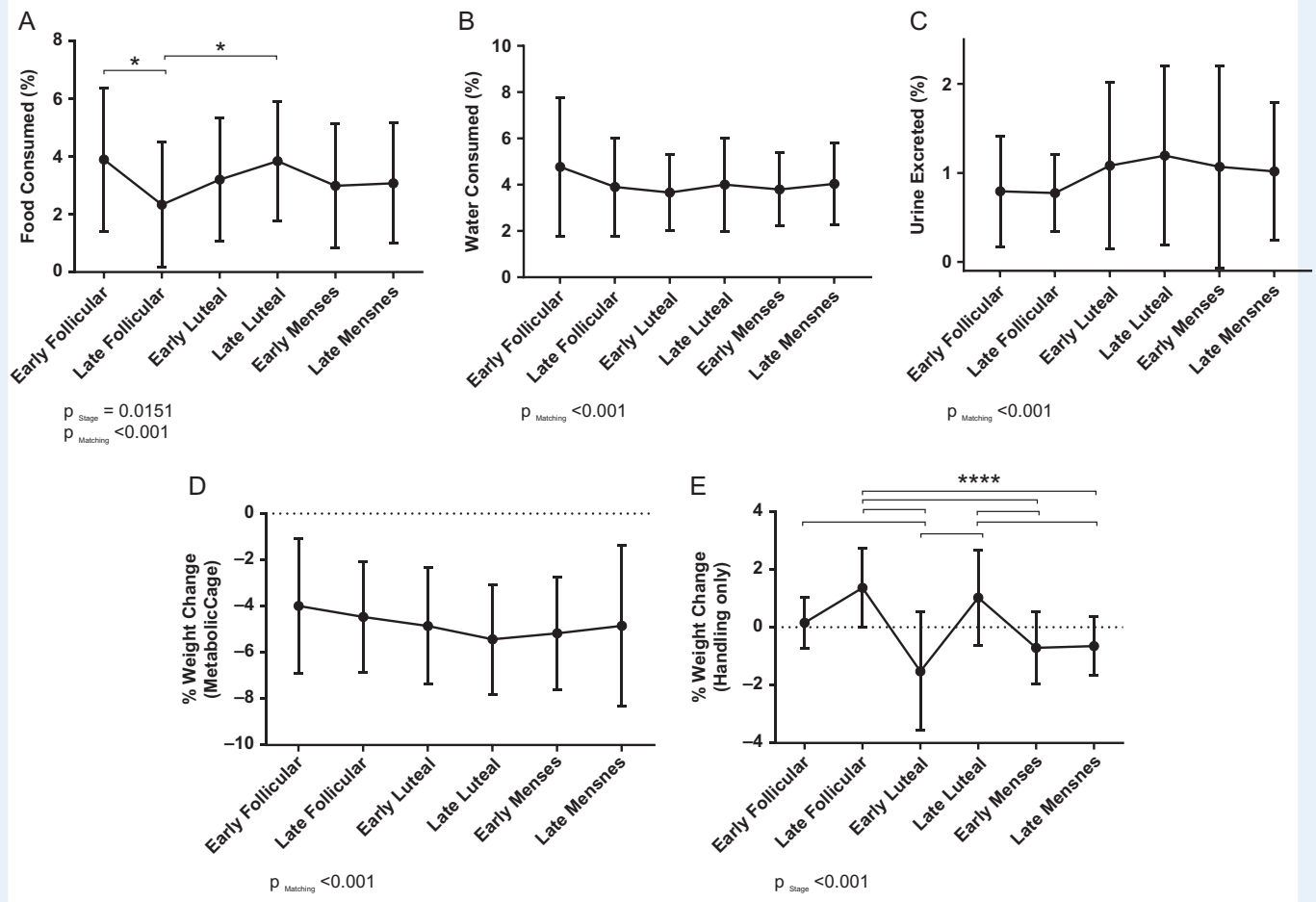
premenstrual spiny mice may seek out a familiar female in preference to a complete stranger, or whether neither animal was sought. In all stages, a preference for interaction with animals was observed rather than isolation in the neutral zone. Our data showed the spiny mice did not show a strong preference to explore the stranger mouse in any phase. However, our social novelty test only looked at other females, and ovariectomised (stranger) or EF phase (known) animals. Given the menstrual cycle of females housed in group cages do not synchronize and therefore each animal was in a different phase of their cycle during the semi-quantitative assessment, there is a possibility that those spiny



**Figure 6** Novel object recognition test analyses across the menstrual cycle in female spiny mice. The novel object recognition test (NORT)1 did not show a preference for either objects (A) and no differences were observed across the menstrual cycle in distance or velocity (B, C). NORT2 (D–F) showed no significant changes in preference for exploration of objects, nor in distance or velocity, however a highly significant effect ( $P < 0.001$ ) of subject matching was found in distance and velocity of both tests (B, C, E, F), ( $n = 11$ ), mean  $\pm$  SD, Repeated measures-ANOVA.



**Figure 7** Social novelty test outcomes across the menstrual cycle in female spiny mice. Social novelty (SN) discrimination index showed no differences were seen across the menstrual cycle in preference to explore either the stranger or known subjects (A) although females consistently preferred exploring other spiny mice (Animal) compared to the Neutral Zone (B), ( $n = 11$ ), mean  $\pm$  SD, Repeated measures-ANOVA.



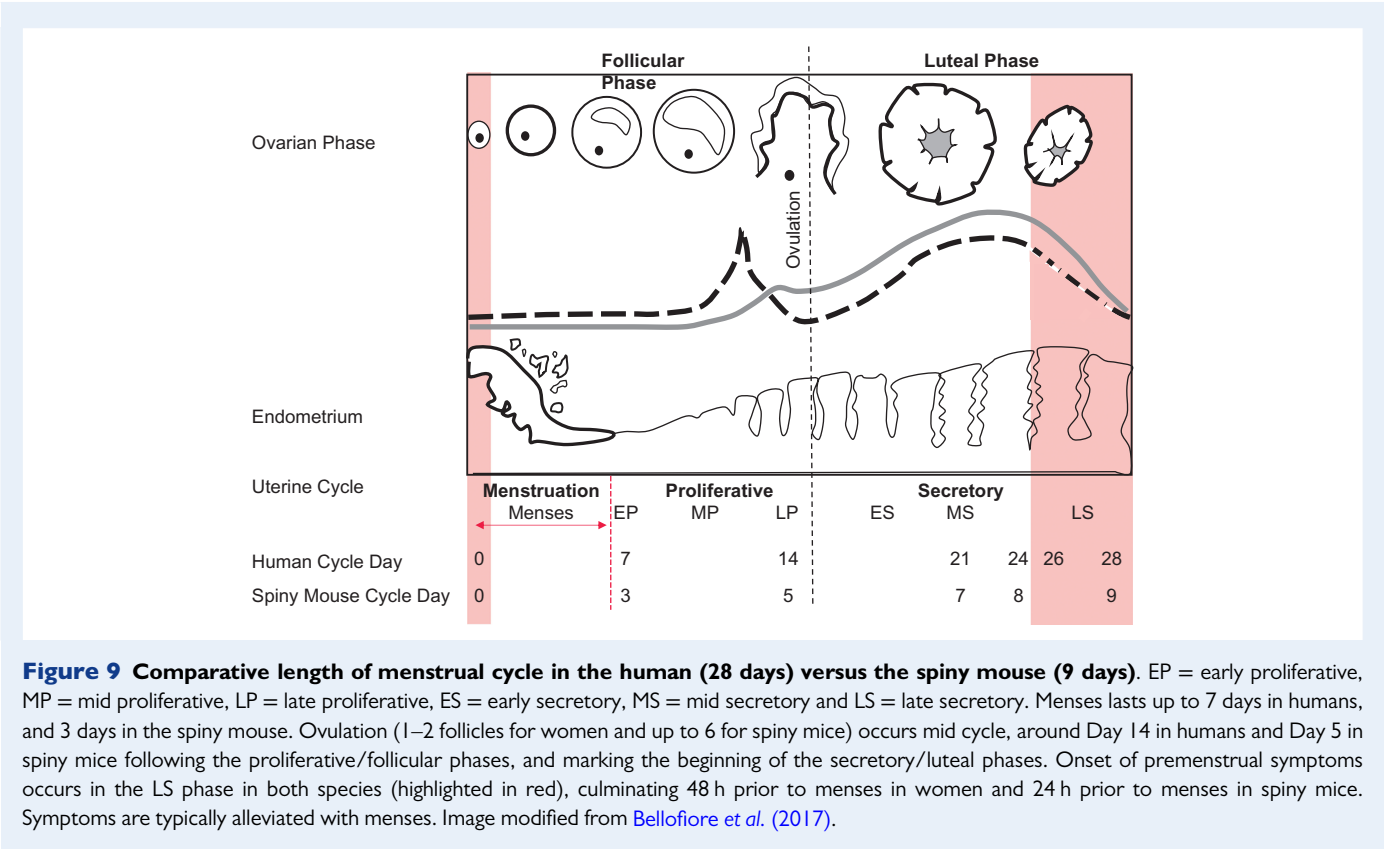
**Figure 8 Metabolic changes across the menstrual cycle in spiny mice.** Females ( $n = 20$ ) show significant peaks in food consumption during the early follicular and late luteal phases compared to the late follicular phase (\*). Water consumption (B) and urine excretion (C) demonstrate a significant effect of subject variation. Changes in body weight after 24 h metabolic cage isolation were negative overall, with no significant variations between menstrual cycle phase but significant effect of subject matching (D). Changes in body weight in conventional housing demonstrate significant fluctuations in each of the phases (E). Data are mean  $\pm$  SD, Repeated measures-ANOVA.

mice which were isolated in the premenstrual phases may have been avoiding interaction with other premenstrual females, more inquisitive females during other phases of their menstrual cycles, or the investigator only. The spiny mouse may prove useful in assessing premenstrual male–female interactions and further studies may expose altered behaviour in the luteal phase.

This novel experimental design of behavioural testing was optimized using ovariectomised females, and no significant differences were noted in any of the primary outcomes for any test. Previous work with mice (Holmes and Rodgers, 1998) demonstrated interesting behavioural adaptations to consecutive exposure to the elevated plus maze ('one trial tolerance'), and increased avoidance to the open arms across 3 days was observed. However, this trial was performed in males only, with inter-trial intervals of only 24 h (compared to 5/6 of our trials at 48 h). Furthermore, ovariectomised controls response to EPM demonstrated no significant differences to entries into either closed/open arms, and we therefore conclude our design is sufficient to negate the effects of repeated exposure. The effect of test  $\times$  test

interaction was not obvious in our ovariectomised cohort, although this may still be a possibility given our controls were of low sample number. However, our method of alternating test order appears sufficient to minimize this in OF, EPM and SI, though this requires further analysis for NORT, given that the unknown object in NORT2, while always preferred in each stage of the menstrual cycle, was not pre-dominant in any stage.

Further similarities in the menstrual cycle between spiny mice and primates are demonstrated in the fluctuations in weight and food consumption. Studies exploring the relationship between eating habits and the menstrual cycle are numerous, with the consensus indicating an increase in appetite, food cravings, basal metabolic rate and energy requirements during the premenstrual phase in women (Solomon *et al.*, 1982; Johnson *et al.*, 1994; Strassmann, 1996; Dye and Blundell, 1997). The spiny mouse housed in group cages peaks in relative weight gain during the late follicular phase (ovulation), and this likely reflects the increased food consumption in the early follicular phase; a pattern observed in many rodents. However, females show a secondary



almost equivalent peak in weight and food consumption during the premenstrual late luteal phase, a trait exemplified by primates (Dye and Blundell, 1997), likely due to the high metabolic demand of spontaneous decidualization (Bellofiore et al., 2018a,b; Strassmann, 1996) and may indicate behaviour analogous to the carbohydrate cravings experienced by premenstrual women (Wurtman et al., 1989). Increased feeding is also seen in premenstrual baboons in conjunction with increased isolation (Hausfater and Skoblick, 1985). No inter-phase differences in water consumption, urine or faeces excretion was observed, although for both these parameters a significant effect of subject matching was noted.

The severity of premenstrual symptoms stems from natural variation of the individual spiny mouse, as when analysing the effect of repeated measures, a continual significant effect of subject matching is observed across each behavioural test. With numerous variants of premenstrual symptoms, there is overwhelming evidence to support that within subject biological variation is a key component in assessing the severity of said symptoms. Progesterone levels, for example, differ not only between women, but within each menstrual cycle of the individual (Jasienska and Jasienski, 2008). The intensity of premenstrual symptoms (behavioural in particular) is also related to gene polymorphisms in estrogen receptors (Huo et al., 2007), dysregulation of serotonin uptake (Rapkin, 1992) and recent evidence suggest an atypical response to acute changing levels (from low to high) of ovarian steroids, with postulated steroid receptor signalling dysregulation (Schmidt et al., 2017). The spiny mouse may enable an in-depth analysis of the behavioural, metabolic, environmental, genetic and endocrinological interplay in the development and exacerbation of PMS and, potentially, PMDD.

We have confirmed the physiological aspects of the menstrual cycle in the spiny mouse and investigated the timing, degree and repair of menstruation (Bellofiore et al., 2017, 2018a, b). The similarities expressed in the biology of this species with our own, strongly suggests correlations in behaviour between the spiny mouse and the human during the premenstrual/early menstrual phases. Identification of anxiety-like behaviour through reduced exploration suggests a state akin to that of PMS in women, and may provide the opportunity to further explore extreme cases of fear and/or withdrawal. These mice with extreme reactions/behavioural manifestations could, in turn, prove highly useful for studies relating to PMDD. As in humans, spiny mice appear to display a spectrum of PMS responses, and these data further highlight the importance of careful behavioural characterization of each mouse to prevent heterogeneity of data in this model.

We provide unique evidence that spiny mice indeed demonstrate cyclical changes in both metabolism and behaviour in different phases of their menstrual cycle; further highlighting the utility of this model for use in multiple aspects of menstrual related research. This study demonstrates a crucial necessity to account for individual variations in both physiological and psychological measurements, as required in humans, further emphasizing the relevance of the spiny mouse as a species in which to study menstruation and associated premenstrual behavioural and metabolic conditions.

Supplementary data

Supplementary data are available at Human Reproduction online.



## Acknowledgements

The authors wish to thank Mr Simon Pederick for technological assistance, and Ms Soo Jin Ahn for assistance in project planning. We also thank Dr Hayley Dickinson for assistance with data compilation and financial provisions for the spiny mice, and Dr Stacey Ellery for assistance with metabolic cage set-up.

## Authors' roles

N.B. designed and performed experiments and analysed the data, F.C. and P.T.-S. provided critical reading of the article, J.E. assisted in interpretation of the data and article drafting.

## Funding

N.B. is supported by a Research Training Program stipend through Monash University. J.E. is supported by a Fellowship awarded by the Fielding Foundation. The Hudson Institute of Medical Research is supported by the Victorian Government Operational Research Infrastructure Support.

## Conflict of interest

The authors declare no conflicts of interest.

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